

PRECICE® Adenosine Kinase (ADK) Assay Kit

includes human recombinant adenosine kinase

For research use only. Not for use in diagnostic procedures

I. Background

Adenosine kinase (ADK) catalyzes the transfer of γ -phosphate from ATP to the 5'-hydroxyl of adenosine, a key modulator of central nervous system functions and a signal molecule involved in hypoxia, inflammation, and nociception. Together with adenosine deaminase, adenosine kinase regulates intra- and extracellular adenosine concentration. Once released outside the cell, adenosine, a naturally occurring ribonucleoside, reveals strong neuroprotective and anti-inflammatory properties. One of the strategies to increase extracellular adenosine consists in inhibiting adenosine kinase (ADK), a ubiquitous enzyme that catalyzes the transfer of γ -phosphate from ATP to the 5'-hydroxyl of adenosine generating AMP and ADP.

Inhibition of adenosine kinase selectively increases local adenosine concentrations and reduces seizure susceptibility and nociception in vivo. ADK dysfunction is implicated in diabetes, epilepsy, and cancer. Consequently, ADK is a rational therapeutic target for drug discovery, and adenosine-regulating drugs have been tested as analgesic, anti-inflammatory, and neuroprotective agents.

II. Principle

PRECICE® Adenosine kinase (ADK) Assay Kit is a non-isotopic assay that allows a continuous monitoring of human recombinant adenosine kinase activity. The assay is developed in microplate format allows a HTS search of novel inhibitors of human recombinant adenosine kinase (ADK) provided with the kit.

NOVOCIB's human adenosine kinase is an active and purified, 345-aa short form (39kDa) ADK protein cloned by RT-PCR amplification of mRNA extracted from human hepatoma cells and expressed in E.coli. The sequence of the cloned ADK (GenBank accession number U50196) was confirmed by DNA sequencing (100% identity). The enzyme is characterized for its substrate properties (Km and Vmax). Its kinetic parameters were compared with published data for natural nucleoside and nucleoside analogues (Table 1).

Table 1. Characterization of the substrate properties (Km and Vmax) of nucleoside and nucleoside analogues for NOVOCIB human adenosine kinase, compared with published kinetic parameters.

Substrate	Km (μ M, Novocib)	Kcat (min^{-1} , Novocib)	Km (μ M, Published)	Kcat (min^{-1} , Published)	Reference
Adenosine	11	1.5	3.2 / 0.150	13	Wu et al. 2005; Yamada et al. 1981
Ribavirin	328	1.9	540	1.8	Wu et al. 2005
Deoxyadenosine	295	3.4	360	-	Yamada et al. 1981
Tubercidine	12	2.2	-	-	-
Inosine	1758	2.6	-	-	-

PRECICE® ADK Assay Kit relies on following coupled enzymatic reactions (Figure 1):

- In the presence of inosine (substrate of ADK) and ATP, ADK phosphorylates inosine to inosine monophosphate (IMP).
 - IMP is immediately oxidized by IMP-dehydrogenase (IMPDH) to XMP, with concomitant formation of NADH_2 .
 - The increase in absorbance at 340 nm directly correlates with ADK activity, allowing continuous measurement of ADK activity
 - In the presence of an inhibitor, inosine phosphorylation is inhibited, resulting in a decrease in NADH_2 formation.

Because IMPDH enzyme is provided in large excess, the coupling reaction proceeds instantly, ensuring that the rate-limiting step is ADK activity. Thus, the enzymatic activity of ADK —corresponding to the formation rate of IMP— is stoichiometrically and directly monitored by measuring NADH₂ production at 340 nm (Fig. 1).

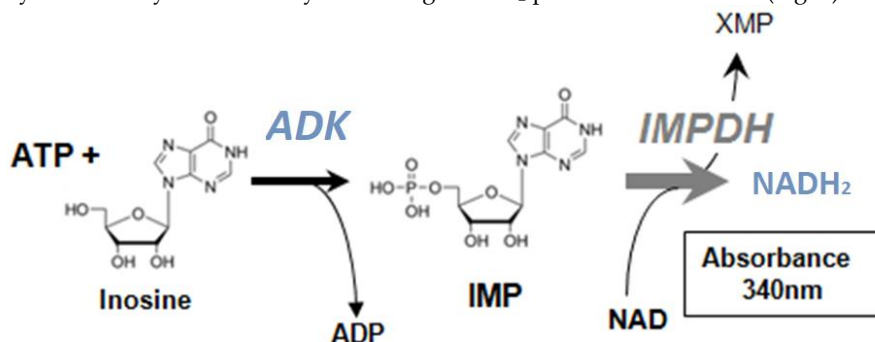


Figure 1. Enzymatic principle of PRECICE® Adenosine Kinase Assay Kit.

PRECICE® ADK Assay Kit was validated with a known ADK inhibitor (A-134974) and meets the requirements of a convenient and reliable HTS assay (microplate format, “add-and measure”, spectrophotometric continuous readout, Z-factor = 0.68).

III. Equipment required

- 1) Plate agitator
- 2) Plate reader fitted with a filter 340nm (ex. Labsystems iEMS Reader MF (Thermo), Epoch (BioTec); PerkinElmer).

IMPORTANT:

The following instructions are given to measure the activity of ADK enzyme *in vitro*, in a range allowing this measurement by spectrophotometry as described here below. NovoCIB does not guarantee the use of its PRECICE® ADK Assay Kit or of one or several of its components, in other conditions than those described in this user manual and/or for other purpose than R&D.

IV. Kit Content:



A standard PRECICE® ADK Phosphorylation Assay Kit (one 96-well plate) contains:

1. one tube "Cofactor 1" (DTT powder);
2. one tube "Cofactor 2" (NAD powder);
3. one tube "Cofactor 3" (ATP powder);
4. one tube "Highly active IMPDH" (lyophilized);
5. one tube "Human recombinant ADK enzyme", (lyophilized, 20nmols);
6. one tube "Inosine" (substrate, 50mM solution);
7. one tube "ADK inhibitor" (A-134974, 50μM solution);
8. one tube "Reaction Buffer 5x", 4ml (provided in 50ml tube)
9. Transparent 96-well plate (round-bottom 96-well plate Corning, Costar®, ref. 3797)
- 10.

V. Preparation of “Reaction buffer”

Important: Spin the tubes before opening to recover their content in the bottom

1. Add 15 ml (corresponding to 15g) of deionized water to provided 50-mL tube with "Reaction Buffer 5x" to prepare “Reaction Buffer 1x”;
2. Add 200μL of deionized water to five following tubes: "Cofactor 1", "Cofactor 2", "Cofactor 3", "Highly active IMPDH" and "Human ADK enzyme". Close, vortex, leave for 5 min, vortex again until complete solubilization of powder.

- Transfer the content of tubes with solubilized "Cofactor 1", "Cofactor 2", "Cofactor 3", "Highly active IMPDH" and "Human ADK enzyme" into a vial "Reaction buffer 1x". Respect the order.

Important: Do not add "Inosine" (substrate)!

Once prepared, complete "Reaction mixture 1x" containing cofactors and enzymes should be used on the same day

VI. Preparation of inhibitor

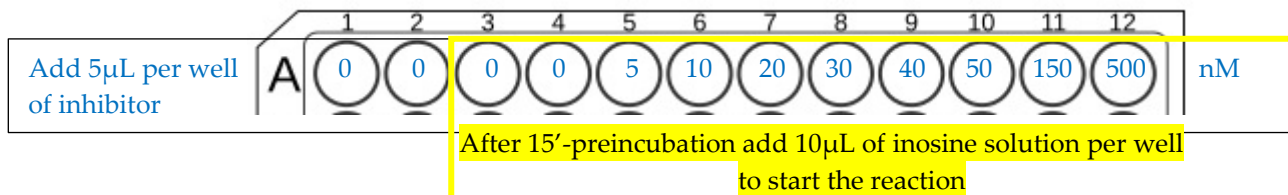
Prepare dilutions for each compound to be tested. Add 5µL per well. Add 5µL of DMSO as negative control

Table 2. Dilution of ADK inhibitor A-134974 (example)

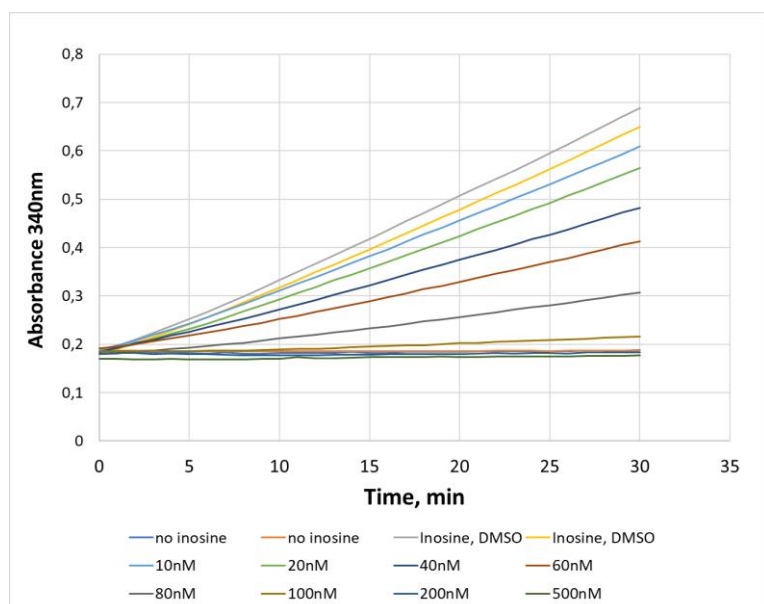
Volume ADK inhibitor, 50µM A-134974, µL	0	10	10	10	10	10	10	10	10
DMSO, µL	100	1240	615	302,5	198	146	105	52,5	15
A-134974 concentration in 40x stock, µM	0	0,4	0,8	1,6	2,4	3,2	4	8	20
Final A-134974 concentration, µM (5µL per well, 200-µL well), nM	0	10	20	40	60	80	100	200	500

VII. Microplate preparation

- Positive** control. Add 5µL of each solution per well (prepared as in **Section VI**);
- Add 5µL of studied compound solution per well;



- Add 200µL of complete "Reaction mixture 1x" containing NAD, DTT, IMPDH, ATP and ADK per well;
- Program plate reader for kinetics absorbance reading (every 1 min), 30°C.
- Preincubation 15'**: Insert the plate into the reader pre-heated at 30°C, agitate for 1min and preincubate the reaction at 30°C for 15min.
- To start the reaction, add 10µL of 50mM inosine solution per well;
- Insert the plate into the reader pre-heated at 30°C, agitate for 1min and monitor the reaction at 340nm at 30°C for 30min with data collection every min.



Typical results obtained with **A-134974** are shown on Table 3 and Figure 3.

Figure 3. Effect of A-134974 on inosine phosphorylation by human adenosine kinase (ADK)

Time course of IMP formation by human recombinant ADK incubated in the absence and presence of increasing concentration of A-134974. After vigorous shaking, the absorbance at 340nm was monitored at 30°C using iEMS microplate reader (Thermo) and round-bottom 96-well microplate (Corning® ref 3797).

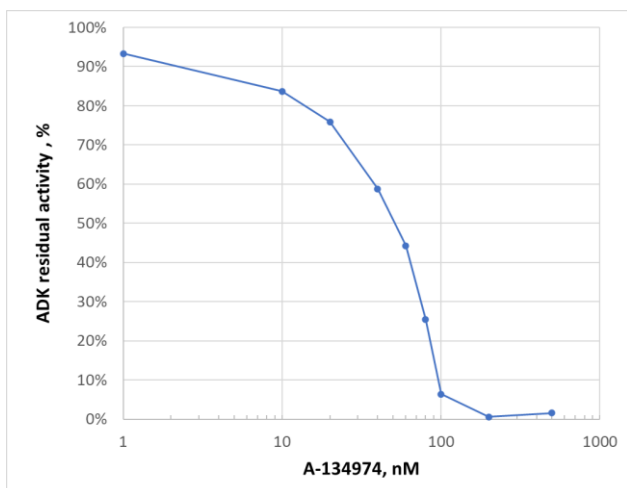


Figure 4. ADK Inhibition curve with compound A-134974 at 8 different concentrations.

Time, min	Absorbance 340 nm											
	no inosine	no inosine	Inosine, DMSO	Inosine, DMSO	10nM	20nM	40nM	60nM	80nM	100nM	200nM	500nM
0	0,183	0,187	0,186	0,179	0,188	0,181	0,184	0,192	0,179	0,185	0,181	0,17
1	0,185	0,187	0,197	0,19	0,198	0,192	0,193	0,195	0,181	0,185	0,182	0,17
2	0,184	0,187	0,21	0,202	0,208	0,2	0,202	0,2	0,185	0,186	0,182	0,169
3	0,183	0,186	0,223	0,215	0,22	0,211	0,209	0,206	0,187	0,187	0,18	0,169
4	0,184	0,186	0,238	0,228	0,231	0,222	0,218	0,212	0,19	0,186	0,181	0,17
5	0,182	0,185	0,252	0,242	0,243	0,232	0,226	0,218	0,193	0,186	0,179	0,169
6	0,181	0,185	0,267	0,256	0,256	0,243	0,235	0,224	0,196	0,187	0,179	0,169
7	0,183	0,185	0,283	0,271	0,27	0,255	0,244	0,231	0,2	0,187	0,178	0,169
8	0,181	0,185	0,299	0,286	0,283	0,268	0,252	0,238	0,203	0,187	0,177	0,169
9	0,181	0,185	0,316	0,302	0,297	0,28	0,262	0,244	0,207	0,188	0,177	0,17
10	0,182	0,186	0,333	0,317	0,311	0,293	0,272	0,252	0,212	0,189	0,177	0,17
11	0,182	0,185	0,35	0,333	0,325	0,306	0,282	0,259	0,216	0,19	0,177	0,173
12	0,182	0,185	0,367	0,349	0,339	0,318	0,291	0,267	0,22	0,19	0,177	0,171
13	0,182	0,186	0,384	0,364	0,353	0,332	0,302	0,274	0,224	0,192	0,178	0,171
14	0,184	0,186	0,401	0,38	0,368	0,344	0,312	0,281	0,228	0,194	0,178	0,172
15	0,182	0,186	0,418	0,396	0,382	0,357	0,322	0,289	0,233	0,195	0,178	0,173
16	0,182	0,185	0,436	0,413	0,396	0,37	0,333	0,297	0,237	0,196	0,179	0,173
17	0,184	0,186	0,454	0,429	0,411	0,384	0,344	0,305	0,242	0,198	0,179	0,174
18	0,184	0,186	0,471	0,446	0,427	0,397	0,354	0,314	0,247	0,198	0,179	0,174
19	0,184	0,186	0,489	0,462	0,441	0,41	0,364	0,321	0,251	0,2	0,18	0,175
20	0,184	0,186	0,507	0,478	0,456	0,424	0,375	0,329	0,256	0,202	0,18	0,174
21	0,185	0,186	0,524	0,495	0,471	0,438	0,385	0,338	0,261	0,203	0,181	0,174
22	0,185	0,187	0,542	0,512	0,486	0,452	0,395	0,346	0,266	0,205	0,182	0,175
23	0,185	0,187	0,559	0,528	0,501	0,465	0,406	0,353	0,272	0,206	0,181	0,175
24	0,186	0,187	0,577	0,545	0,516	0,479	0,417	0,362	0,277	0,208	0,182	0,175
25	0,186	0,186	0,595	0,562	0,53	0,492	0,426	0,37	0,28	0,209	0,182	0,175
26	0,186	0,187	0,613	0,579	0,546	0,507	0,437	0,378	0,285	0,21	0,181	0,175
27	0,187	0,187	0,632	0,597	0,562	0,521	0,449	0,387	0,291	0,211	0,183	0,176
28	0,187	0,187	0,651	0,614	0,577	0,535	0,46	0,396	0,296	0,213	0,183	0,176
29	0,187	0,187	0,67	0,632	0,593	0,55	0,472	0,405	0,302	0,215	0,183	0,176
30	0,188	0,187	0,689	0,649	0,609	0,564	0,482	0,413	0,307	0,216	0,183	0,177
Rate, AU/min	no inosine	no inosine	Inosine, DMSO	Inosine, DMSO	10nM	20nM	40nM	60nM	80nM	100nM	200nM	500nM
	0,000	0,000	0,017	0,016	0,014	0,013	0,010	0,008	0,004	0,001	0,000	0,000
IMP formation, % of control			100%	93%	84%	76%	59%	44%	26%	6%	1%	2%